

Time Spent: 300
Subject:
General Chemistry

QID: 402400
System:
Atomic Theory and Chemical Composition

$^{18}\text{F}^- + \text{H}_2\text{O} \rightarrow ^{18}\text{F}^- + 2\text{H}^+ + \frac{1}{2}\text{O}_2$

Ion exchange column

K_2CO_3
 $R = 2.2.2\text{-Cryand}$
 CH_3CN

$[\text{R}]\text{K}^+ ^{18}\text{F}^-$

Mannose tritiate

$\text{CH}_3\text{C}\equiv\text{S}=\text{Ac}$ $\text{CF}_3\text{SO}_2\text{S}=\text{TF}$

$[\text{R}]\text{K}^+ ^{18}\text{F}^-$
 CH_3CN

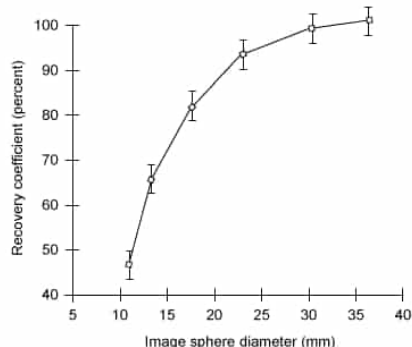
1. NaOH
 2. HCl

2-deoxy-2-(^{18}F)
 fluoro-D-glucose

Initially, a target of ^{18}O -enriched water ($^{18}\text{OH}_2$) is bombarded by protons using a cyclotron to achieve a (p-n) nuclear reaction in which a proton hitting the ^{18}O nucleus is absorbed by the nucleus, causing the ejection of a neutron and forming an ^{18}F nucleus. Following separation of the resulting $^{18}\text{F}^-$ anions from the residual $^{18}\text{OH}_2$ via an ion exchange column, $^{18}\text{F}^-$ is used in a nucleophilic substitution reaction with an acetylated mannose triflate substrate. Subsequent hydrolysis to remove the acetyl protecting groups yields ^{18}F -FDG.

Time (min)	¹⁸ F Sample remaining (percent)
0	100
15	88
30	78
45	68
60	60
75	53
90	47
105	42
120	37
135	33
150	29
165	26
180	23
195	21
210	19
225	18
240	17

Because of limitations in the spatial resolution of PET scanners, when an object (such as a tumor) is smaller than 30 mm in diameter, the measured specific uptake value (SUV) of ^{18}F -FDG accumulated within the object is lower than the actual SUV. The recovery coefficient (the ratio of the measured SUV to the actual SUV) shows that the measured SUV decreases as the size of the object decreases below the 30-mm threshold (Figure 3).



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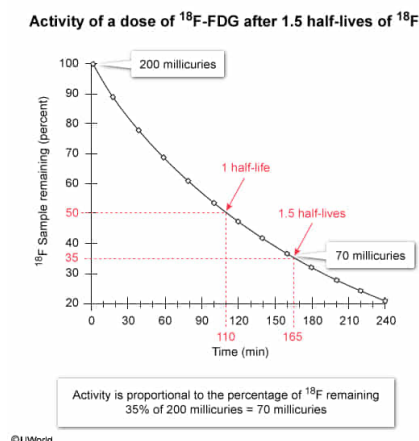
If the initial activity from a 1-mL dose of ^{18}F -FDG is 200 millicuries, what will be the approximate activity of the dose after 1.5 half-lives of ^{18}F have passed?

- ☐ A. 20 millicuries [2%]
☐ B. 50 millicuries [15%]
☒ C. 70 millicuries [79%]
☐ D. 100 millicuries [1%]

Correct

79% answered correctly

Explanation:



In **radioactive decay**, the **half-life** of a radioisotope is the time required for the amount of the isotope to **decrease by half** (50% of the initial amount). With each subsequent half-life interval that passes, the amount of the isotope that remains is again decreased by half such that the fraction of the initial amount that remains becomes **exponentially smaller** and eventually approaches zero. Accordingly, the fraction remaining after n half-lives can be calculated as:

$$\text{Fraction remaining} = \frac{1}{2^n}$$

The activity of a radioisotope refers to the number of decay events per unit of time. An older unit of activity called the curie (Ci) was initially developed based on the radium-226 isotope, where 1 curie is equal to 3.7×10^{10} radioactive decays per second, but in the International System of units (SI), the preferred unit of activity is the becquerel (Bq), where 1 Bq is equal to 1 radioactive decay per second. As such, radioisotope activity is proportional to the amount of the decaying radioisotope present, and activity decreases by 50% with each successive half-life that passes.

$$\text{Measured activity} = (\text{Fraction remaining}) \times (\text{Initial activity})$$

$$\text{Measured activity} = \frac{1}{2^n} \times (\text{Initial activity})$$

Based on the nuclear decay profile of the ^{18}F radionuclide shown in Figure 2 of the passage, the half-life (point of 50% decay) for ^{18}F is 110 minutes; therefore, 1.5 half-lives equal 165 minutes. After 165 minutes, only 35% of the initial amount of ^{18}F would remain. As a result, the activity for the sample would also decrease to 35% of the initial activity, resulting in a measured activity of 70 millicuries.

$$\text{Measured activity} = (\text{Fraction remaining}) \times (\text{Initial activity})$$

$$70 \text{ millicuries} = 0.35 \times 200 \text{ millicuries}$$

(Choice A) An activity of 20 millicuries would be seen after a little more than *three half-lives* have elapsed.

(Choice B) An activity of 50 millicuries would be measured after *two half-lives* have elapsed.

(Choice D) An activity of 100 millicuries would be detected after *one half-life* has elapsed.

Educational objective:

The activity of a radioisotope (the number of decay events per unit of time) is proportional to the amount of the decaying radioisotope present, which decreases by 50% for each radioisotope half-life that passes.

Time spent:0 sec
Subject:
General Chemistry

QID:402459
System:
Atomic Theory and Chemical Composition

The radiopharmaceutical 2-deoxy-2-(^{18}F)fluoro-D-glucose (abbreviated as ^{18}F -FDG) is a modified form of D-glucose in which the hydroxyl group on the 2-carbon of the molecule has been replaced with a positron-emitting ^{18}F radionuclide. Synthesis of ^{18}F -FDG is accomplished in several steps involving both nuclear and chemical reactions, as shown in Figure 1.

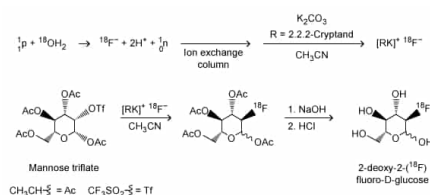


Figure 1 Synthetic scheme for 2-deoxy-2-(^{18}F)fluoro-D-glucose

Initially, a target of ^{18}O -enriched water ($^{18}\text{OH}_2$) is bombarded by protons using a cyclotron to achieve a (p-n) nuclear reaction in which a proton hitting the ^{18}O nucleus is absorbed by the nucleus, causing the ejection of a neutron and forming an ^{18}F nucleus. Following separation of the resulting $^{18}\text{F}^-$ anions from the residual $^{18}\text{OH}_2$ via an ion exchange column, $^{18}\text{F}^-$ is used in a nucleophilic substitution reaction with an acetylated mannose triflate substrate. Subsequent hydrolysis to remove the acetyl protecting groups yields ^{18}F -FDG.

Because ^{18}F -FDG is an analog of D-glucose, it is readily taken into cells by the same metabolic processes as D-glucose. Once inside the cells, the positron emissions of the ^{18}F radionuclide can be detected by positron emission tomography (PET) scans to produce diagnostic medical images of objects such as tumors. Emissions from the ^{18}F radionuclide subsequently diminish over time via radioactive decay (Figure 2).

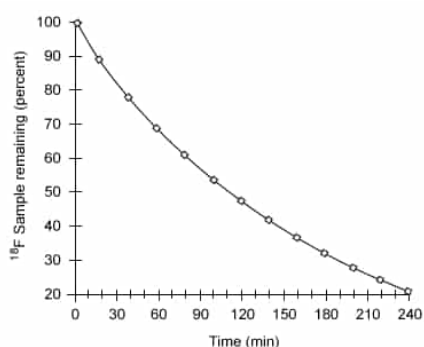


Figure 2 Nuclear decay of the ^{18}F radionuclide

Because of limitations in the spatial resolution of PET scanners, when an object (such as a tumor) is smaller than 30 mm in diameter, the measured specific uptake value (SUV) of ^{18}F -FDG accumulated within the object is lower than the actual SUV. The recovery coefficient (the ratio of the measured SUV to the actual SUV) shows that the measured SUV decreases as the size of the object decreases below the 30-mm threshold (Figure 3).

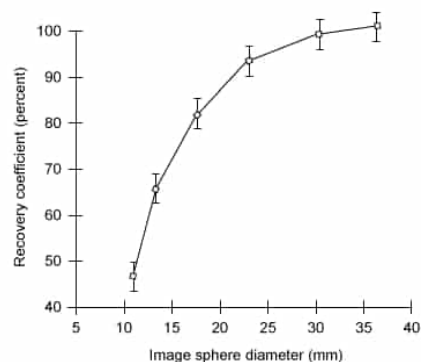


Figure 3 Recovery coefficient as a function of object size in a PET scan with a spatial resolution of 7 mm using ordered subsets expectation maximization (OSEM) image processing

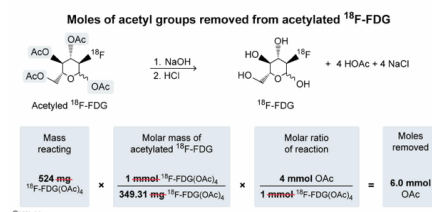
The last step in the synthesis of ^{18}F -FDG is a hydrolysis reaction to remove the four acetyl (OAc) protecting groups. If 524 mg of acetylated ^{18}F -FDG is hydrolyzed, how many millimoles of the acetyl group are removed in the reaction? (Note: The molar mass of the acetylated ^{18}F -FDG is $349.31 \text{ g/mol} = 349.31 \text{ mg/mmol}$.)

- ☐ A. 0.4 mmol [6%]
☐ B. 1.5 mmol [35%]
☒ C. 6.0 mmol [54%]
☐ D. 7.5 mmol [3%]

Correct

53% answered correctly

Explanation:



The **molar mass** of a compound is the amount of mass contained in 1 mole of the compound (grams per mole) and can be used to convert the **mass** of a compound to the number of **moles**. The reactants and/or products of a particular reaction are related to one another by the **mole ratios** from the **balanced reaction equation** or chemical formula stoichiometry.

If 524 mg of ^{18}F -FDG(OAc)₄ (acetylated ^{18}F -FDG) participates in a hydrolysis reaction to remove the 4 acetyl groups as shown in Figure 1, the number of moles of ^{18}F -FDG(OAc)₄ present in the reaction can be calculated by dividing the given mass of ^{18}F -FDG(OAc)₄ by its molar mass, which gives 1.5 mmol of ^{18}F -FDG(OAc)₄.

Comparing the structure of ^{18}F -FDG(OAc)₄ with ^{18}F -FDG shows that 4 moles of acetyl groups (OAc) are removed for every 1 mole of ^{18}F -FDG(OAc)₄ that reacts. Applying this ratio, the number of moles of acetyl group removed can be found by multiplying the millimoles of ^{18}F -FDG(OAc)₄ by 4. Therefore, 6.0 mmol of the acetyl group are removed.

(Choice A) A quantity of 0.4 mmol is obtained if the ratio of 4 acetyl groups per 1 acetylated ^{18}F -FDG molecule is inverted during the calculation.

(Choice B) There are 1.5 mmol of ^{18}F -FDG(OAc)₄ molecules in the given mass but there are four times as many millimoles of acetyl groups removed from the ^{18}F -FDG(OAc)₄ molecules during the reaction (ie, each mole of ^{18}F -FDG(OAc)₄ releases 4 moles of OAc).

(Choice D) A value of 7.5 mmol is obtained if a 5:1 mole ratio is used in the calculation, but the hydrolysis step of the reaction pertains to the removal of only 4 acetyl groups per hydrolyzed molecule.

Educational objective:

The number of moles of a compound is found by dividing the mass of the sample by its molar mass. Stoichiometric mole ratios taken from a balanced reaction equation or a chemical structure can be applied to determine the number of moles of one chemical species relative to the moles of another chemical species in the reaction.

Time spent: 0 sec
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 General Chemistry

QID: 402460
 System:
 Atomic Theory and Chemical Composition

$^{18}\text{F}^- + \text{H}_2\text{O} \rightarrow ^{18}\text{F}^- + 2\text{H}^+ + \frac{1}{2}\text{O}_2$

Ion exchange column

K_2CO_3
 $R = 2.2.2\text{-Crylodan}$
 CH_3CN

$[\text{R}K]^{18}\text{F}$

$[\text{R}K]^{18}\text{F}^-$
 CH_3CN

1. NaOH
 2. HCl

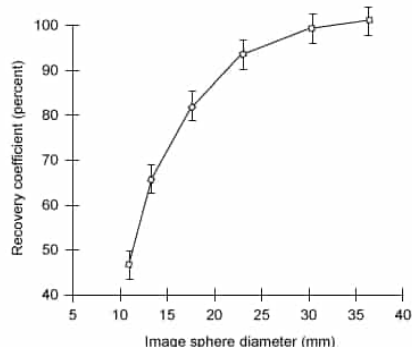
Mannose tritiate
 $\text{CH}_3\text{C}(\text{O})\text{SH} = \text{Ac}$ $\text{CF}_3\text{SO}_2\text{S} = \text{Tf}$

2-deoxy-2-(^{18}F)
 fluoro-D-glucose

Initially, a target of ^{18}O -enriched water ($^{18}\text{OH}_2$) is bombarded by protons using a cyclotron to achieve a (p-n) nuclear reaction in which a proton hitting the ^{18}O nucleus is absorbed by the nucleus, causing the ejection of a neutron and forming an ^{18}F nucleus. Following separation of the resulting $^{18}\text{F}^-$ anions from the residual $^{18}\text{OH}_2$ via an ion exchange column, $^{18}\text{F}^-$ is used in a nucleophilic substitution reaction with an acetylated mannose triflate substrate. Subsequent hydrolysis to remove the acetyl protecting groups yields ^{18}F -FDG.

Time (min)	¹⁸ F Sample remaining (percent)
0	100
15	89
30	77
45	68
60	61
75	54
90	48
105	43
120	38
135	34
150	31
165	28
180	26
195	24
210	22
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240	20

Because of limitations in the spatial resolution of PET scanners, when an object (such as a tumor) is smaller than 30 mm in diameter, the measured specific uptake value (SUV) of ^{18}F -FDG accumulated within the object is lower than the actual SUV. The recovery coefficient (the ratio of the measured SUV to the actual SUV) shows that the measured SUV decreases as the size of the object decreases below the 30-mm threshold (Figure 3).



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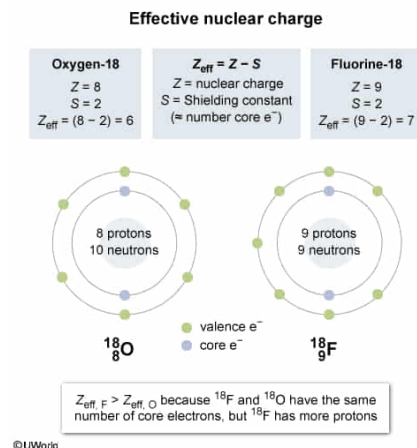
Compared to the effective nuclear charge of ^{18}O , the effective nuclear charge of ^{18}F is:

- ☐ A. higher, because ^{18}F and ^{18}O have the same number of core electrons but ^{18}F has fewer neutrons. [7%]
- ☐ B. lower, because ^{18}F has more valence electrons than ^{18}O . [15%]
- ✓ ☒ C. higher, because ^{18}F and ^{18}O have the same number of core electrons but ^{18}F has more protons. [74%]
- ☐ D. lower, because ^{18}F has an equal number of protons and electrons. [2%]

Correct

74% answered correctly

Explanation:



Each proton in the nucleus of an atom contributes one unit of positive charge that exerts an **electrostatic attraction** on the negatively charged electrons around the atom. However, in atoms with several electrons, the **core electrons** positioned between the valence electrons and the nucleus provide a **shielding constant S** that counteracts part of the

full **nuclear charge Z** attracting the valence electrons. As a result, the **valence electrons** experience an **effective nuclear charge Z_{eff}** that is less than the full nuclear charge, such that

$$Z_{\text{eff}} = Z - S$$

Precisely calculating S is complex and depends upon the electron density of a given electron orbital configuration, but a rough, first-order approximation can be made by setting S equal to the number of **core electrons**.

$$S \approx \text{Core electrons} = \text{Total electrons} - \text{Valence electrons}$$

When comparing atoms within the *same period* (row) of the periodic table, this approximation shows that Z_{eff} increases as the **atomic number** increases, but this approximation does not distinguish between *different rows* of the periodic table and only shows that Z_{eff} increases as the group (column) number increases.

Although ^{18}F and ^{18}O both have two core electrons, ^{18}F has more protons. Therefore, when comparing the effective nuclear charge of ^{18}O to the effective nuclear charge of ^{18}F , the ^{18}F atom has the greater atomic number and higher Z_{eff} :

$$Z_{\text{eff, O}} = Z_{\text{O}} - S_{\text{O}} = (8 - 2) = 6$$

$$Z_{\text{eff, F}} = Z_{\text{F}} - S_{\text{F}} = (9 - 2) = 7$$

(Choice A) Neutrons have no charge and do not contribute to the nuclear charge of an atom.

(Choice B) Although valence electrons can contribute slightly to the shielding constant S , nuclear protons and **core electrons** influence Z_{eff} to a much greater extent than valence electrons.

(Choice D) All atoms in their elemental state have equal numbers of protons and electrons, but the difference in Z_{eff} between two atoms of different elements depends on the relative number of nuclear protons and core electrons.

Educational objective:

The nuclear charge Z (number of protons) influencing a valence electron is diminished by the shielding constant S , which results in an effective nuclear charge Z_{eff} that is less than the full nuclear charge. By setting S equal to the

number of core electrons, Z_{eff} can be estimated as $Z_{\text{eff}} = Z - S$.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402461

System:
Atomic Theory and Chemical Composition

$I_2 + 18O_2H_2 \rightarrow {}^{18}F^- + 2H^+ + I_2^-$

$[R]K^+ {}^{18}F^- \xrightarrow[\text{CH}_3\text{CN}]{\text{Ion exchange column}} [RK]^+ {}^{18}F^-$

$R = 2,2,2\text{-}^3\text{Cryptand}$

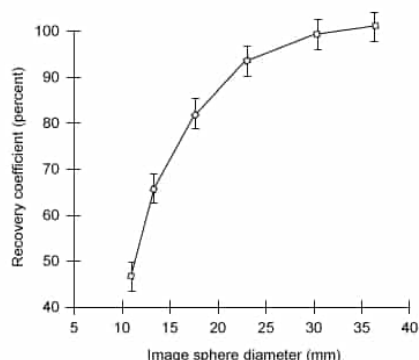
Mannose triflate
 $CH_3CH-S = Ac$ $CF_3SO_2-S = Tf$

2-deoxy-2-(^{18}F) fluoro-D-glucose

Initially, a target of ^{18}O -enriched water ($^{18}\text{OH}_2$) is bombarded by protons using a cyclotron to achieve a (p-n) nuclear reaction in which a proton hitting the ^{18}O nucleus is absorbed by the nucleus, causing the ejection of a neutron and forming an ^{18}F nucleus. Following separation of the resulting $^{18}\text{F}^-$ anions from the residual $^{18}\text{OH}_2$ via an ion exchange column, $^{18}\text{F}^-$ is used in a nucleophilic substitution reaction with an acetylated mannose triflate substrate. Subsequent hydrolysis to remove the acetyl protecting groups yields ^{18}F -FDG.

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210	20

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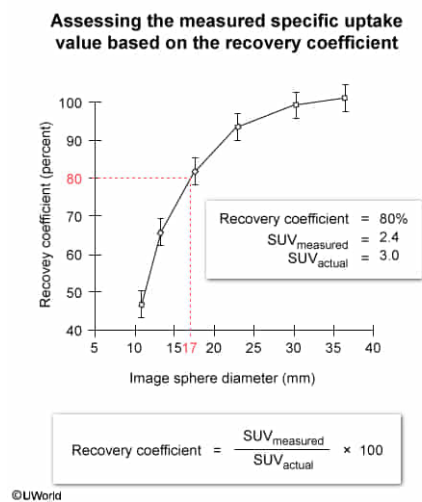
Based on the imaging resolution data provided in the passage, what measured specific uptake value (SUV) is likely to be observed when imaging a 1.7-cm tumor with an actual SUV of 3.0?

- ☐ A. 0.6 [8%]
☒ B. 2.4 [72%]
☐ C. 3.0 [12%]
☐ D. 3.6 [5%]

Correct

71% answered correctly

Explanation:



As stated in the passage, limitations in the spatial resolution of PET scanners can cause objects (eg, tumors) with a diameter less than 30 mm to appear smaller than the actual size because the PET scan does not detect all of the ^{18}F -FDG present in the localized tissue.

To quantify this limitation, an analysis of objects with a known size can be performed to measure the recovery coefficient. The passage states that the recovery coefficient is the ratio of the measured SUV to the actual SUV. In Figure 3, this ratio is expressed as a percentage as given by

$$\text{Recovery coefficient} = \frac{SUV_{\text{measured}}}{SUV_{\text{actual}}} \times 100$$

where SUV_{actual} is the amount of ^{18}F -FDG actually taken into the object and SUV_{measured} is the amount of ^{18}F -FDG detected within the object by a scan. As such, full detection is indicated by a recovery coefficient of 100%. Figure 3 shows that the percentage of SUV_{actual} detected decreases as the size of the object decreases below the 30-mm threshold.

Based on the data from Figure 3, a tumor that is 1.7 cm (17 mm) in diameter can be expected to have a recovery coefficient of 80%. If the amount of ^{18}F -FDG present in the tissue is such that $SUV_{\text{actual}} = 3.0$, the expected SUV_{measured} can be calculated using the recovery coefficient.

$$0.80 = \frac{SUV_{\text{measured}}}{SUV_{\text{actual}}}$$

$$0.80 = \frac{SUV_{\text{measured}}}{3.0}$$

$$SUV_{\text{measured}} = 0.80(3.0) = 2.4$$

Therefore, a tumor with a 1.7-cm diameter and an ^{18}F -FDG SUV_{actual} value of 3.0 will have an SUV_{measured} of 2.4.

(Choice A) A quantity of 0.6 (20%) is the amount of the actual SUV that is not detected by the scanner due to resolution limitations.

(Choice C) A measured SUV of 3.0 would be possible only if 100% of the actual SUV were detected, but due to scanner limitations objects with a diameter less than 3.0 cm (30 mm) have a recovery coefficient below 100%.

(Choice D) If an SUV of 3.6 were measured, this would indicate that 20% more than the actual SUV was detected. The measured value should not exceed 3.0, and objects with a diameter less than 30 mm should detect less than this amount.

Educational objective:

Data plots that correlate different measurements can be used to estimate the expected result for similar measurements under the same conditions.

Time spent: 0 sec
Subject:
General Chemistry

QID: 402462
System:
Atomic Theory and Chemical Composition